

Supplements:

Differentially Private Selection using Enhanced Sensitivity Concepts

This document provides the proofs and omitted definitions in the main paper. It also provides the detailed explanations of computation used in the experiments.

I. PROOFS AND OMITTED DEFINITIONS

Theorem 1. M_{ELP} satisfies ϵ -differential privacy.

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{ELP}(x) = 1] \leq e^\epsilon \cdot \Pr[M_{ELP}(y) = 1]$. We denote $Z_r = \text{Lap}_r\left(\frac{EGS_{u, \mathcal{R}}}{\epsilon}\right)$.

$$\begin{aligned} \Pr[M_{ELP}(x) = 1] &= \int_{v \in (-\infty, \infty)} \Pr[u(x, 1) + Z_1 = v] \\ &\quad \cdot \Pr[u(x, 2) + Z_2 \leq v] \\ &\quad \cdots \Pr[u(x, m) + Z_m \leq v] \\ &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(x, 1)] \\ &\quad \cdot \Pr[Z_2 \leq v - u(x, 2)] \\ &\quad \cdots \Pr[Z_m \leq v - u(x, m)]. \end{aligned}$$

$$\begin{aligned} \Pr[M_{ELP}(y) = 1] &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1)] \\ &\quad \cdot \Pr[Z_2 \leq v - u(y, 2)] \\ &\quad \cdots \Pr[Z_m \leq v - u(y, m)] \\ &\geq \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1)] \\ &\quad \cdot \Pr[Z_2 \leq v - u(x, 2) - T] \\ &\quad \cdots \Pr[Z_m \leq v - u(x, m) - T] \\ &\quad \left(T = \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x, r') - u(y, r')| \right) \\ &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1) + T] \\ &\quad \cdot \Pr[Z_2 \leq v - u(x, 2)] \\ &\quad \cdots \Pr[Z_m \leq v - u(x, m)]. \end{aligned}$$

Here, $|u(x, 1) - u(y, 1) + T|$

$$\begin{aligned} &= \left| u(x, 1) - u(y, 1) + \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x, r') - u(y, r')| \right| \\ &\leq |u(x, 1) - u(y, 1)| + \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x, r') - u(y, r')| \\ &\leq \max_{r \in \mathcal{R}} \max_{x, y: d(x, y)=1} \left(|u(x, r) - u(y, r)| \right. \\ &\quad \left. + \max_{r' \in \mathcal{R} \setminus \{r\}} |u(x, r') - u(y, r')| \right) = EGS_{u, \mathcal{R}}. \end{aligned}$$

Therefore, $\forall v : \Pr[Z_1 = v - u(x, 1)] \leq e^\epsilon \cdot \Pr[Z_1 = v - u(y, 1) + T]$, and $\Pr[M_{ELP}(x) = 1] \leq e^\epsilon \cdot \Pr[M_{ELP}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{ELP}(x) = r] \leq e^\epsilon \cdot \Pr[M_{ELP}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Lemma 1. $EGS_{u, \mathcal{R}} \leq 2 \cdot GS_{u, \mathcal{R}}$.

Proof.

$$\begin{aligned} EGS_{u, \mathcal{R}} &= \max_{r \in \mathcal{R}} \max_{x, y: d(x, y)=1} \left(|u(x, r) - u(y, r)| \right. \\ &\quad \left. + \max_{r' \in \mathcal{R} \setminus \{r\}} |u(x, r') - u(y, r')| \right) \\ &\leq \max_{r \in \mathcal{R}} \max_{x, y: d(x, y)=1} \left(\max_{r' \in \mathcal{R}} |u(x, r') - u(y, r')| \right. \\ &\quad \left. + \max_{r' \in \mathcal{R}} |u(x, r') - u(y, r')| \right) \\ &= 2 \cdot \max_{r \in \mathcal{R}} \max_{x, y: d(x, y)=1} |u(x, r) - u(y, r)| \\ &= 2 \cdot GS_{u, \mathcal{R}}. \end{aligned}$$

$\square$

Theorem 2. The $ESPS$ satisfies $\left(\left(k + \frac{|\mathcal{R}|}{2} \cdot l \right) \cdot \epsilon \right)$ -differential privacy.

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{ESPS}(x) = 1] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{ESPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x, r)$.

$$\begin{aligned} \Pr[M_{ESPS}(x) = 1] &= \int_{v \in (-\infty, \infty)} \Pr \left[u(x, 1) + \frac{S(x)}{2\alpha'} \cdot Z_1 = v \right] \\ &\quad \cdot \Pr \left[u(x, 2) + \frac{S(x)}{2\alpha'} \cdot Z_2 \leq v \right] \\ &\quad \cdots \Pr \left[u(x, m) + \frac{S(x)}{2\alpha'} \cdot Z_m \leq v \right] \\ &= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\ &\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2))}{S(x)} \right] \\ &\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m))}{S(x)} \right]. \end{aligned}$$

$$\begin{aligned}
& \Pr[M_{ESPS}(y) = 1] \\
&= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1))}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(y, 2))}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(y, m))}{S(y)} \right] \\
&\geq \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1))}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2) - T)}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m) - T)}{S(y)} \right] \\
&\quad \left(T = \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x, r') - u(y, r')| \right) \\
&= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T)}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2))}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m))}{S(y)} \right].
\end{aligned}$$

Here, for all $v \in (-\infty, \infty)$

$$\begin{aligned}
\forall r' \in \mathcal{R} \setminus \{1\}: \quad & \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(x)} \right] \\
& \leq e^{\frac{l\epsilon}{2}} \cdot \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(y)} \right]
\end{aligned}$$

because

$$\begin{aligned}
\frac{S(x)}{S(y)} &= \frac{\max_r S(x, r)}{\max_r S(y, r)} \leq e^{\beta'} = e^{\beta(l\epsilon)} \\
\text{and } \frac{S(x)}{S(y)} &\geq e^{-\beta'} = e^{-\beta(l\epsilon)}.
\end{aligned}$$

In addition,

$$\begin{aligned}
& \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\
& \leq e^{k\epsilon} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T)}{S(x)} \right] \quad (1)
\end{aligned}$$

because

$$\begin{aligned}
& \frac{|2\alpha'(u(x, 1) - u(y, 1) + T)|}{S(x)} \\
&= \frac{2\alpha' |u(x, 1) - u(y, 1) + \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x, r') - u(y, r')||}{S(x)} \\
&\leq \frac{2\alpha' (|u(x, 1) - u(y, 1)| + \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x, r') - u(y, r')|)}{S(x)} \\
&\leq \frac{2\alpha' \cdot ELS_{u, \mathcal{R}}(x, 1)}{S(x)} \leq \frac{2\alpha' \cdot S(x, 1)}{S(x)} \leq 2\alpha' = \alpha(2k\epsilon),
\end{aligned}$$

and

$$(1) \leq e^{k\epsilon} \cdot e^{\frac{l\epsilon}{2}} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T)}{S(y)} \right].$$

Therefore, $\Pr[M_{ESPS}(x) = 1] \leq \exp((k + \frac{m}{2} \cdot l) \cdot \epsilon) \cdot \Pr[M_{ESPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{ESPS}(x) = r] \leq \exp((k + \frac{m}{2} \cdot l) \cdot \epsilon) \cdot \Pr[M_{ESPS}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Lemma 2. $\forall x \in D^n, \forall r \in \mathcal{R} : ELS_{u, \mathcal{R}}(x, r) \leq 2 \cdot \max_{r' \in \mathcal{R}} LS_{u, \mathcal{R}}(x, r')$.

Proof.

$$\begin{aligned}
ELS_{u, \mathcal{R}}(x, r) &= \max_{y: d(x, y)=1} \left(|u(x, r) - u(y, r)| \right. \\
&\quad \left. + \max_{r' \in \mathcal{R} \setminus \{r\}} |u(x, r') - u(y, r')| \right) \\
&\leq 2 \cdot \max_{r' \in \mathcal{R}} \max_{y: d(x, y)=1} |u(x, r') - u(y, r')| \\
&= 2 \cdot \max_{r' \in \mathcal{R}} LS_{u, \mathcal{R}}(x, r'). \quad \square
\end{aligned}$$

Theorem 3. M_{EDLP} satisfies ϵ -differential privacy.

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{EDLP}(x) = 1] \leq e^\epsilon \cdot \Pr[M_{EDLP}(y) = 1]$. We denote $Z_r = \text{Lap}_r\left(\frac{EDGS_{u, \mathcal{R}}}{\epsilon}\right)$.

$$\begin{aligned}
\Pr[M_{EDLP}(x) = 1] &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(x, 1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x, 2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(x, m)].
\end{aligned}$$

$$\begin{aligned}
\Pr[M_{EDLP}(y) = 1] &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(y, 2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(y, m)] \\
&\geq \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x, 2) - T'] \\
&\quad \cdots \Pr[Z_m \leq v - u(x, m) - T'] \\
&\quad \left(T' = \max_{r' \in \mathcal{R} \setminus \{1\}} (u(y, r') - u(x, r')) \right) \\
&= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1) + T'] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x, 2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(x, m)].
\end{aligned}$$

Here, $|u(x, 1) - u(y, 1) + T'|$

$$\begin{aligned}
&= \left| u(x, 1) - u(y, 1) + \max_{r' \in \mathcal{R} \setminus \{1\}} (u(y, r') - u(x, r')) \right| \\
&\leq \max_{r \in \mathcal{R}} \max_{x, y: d(x, y)=1} \left| (u(x, r) - u(y, r)) \right. \\
&\quad \left. + \max_{r' \in \mathcal{R} \setminus \{r\}} (u(y, r') - u(x, r')) \right| \\
&= EDGS_{u, \mathcal{R}}.
\end{aligned}$$

Therefore, $\forall v : \Pr[Z_1 = v - u(x, 1)] \leq e^\epsilon \cdot \Pr[Z_1 = v - u(y, 1) + T']$, and $\Pr[M_{EDLP}(x) = 1] \leq e^\epsilon \cdot \Pr[M_{EDLP}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{EDLP}(x) = r] \leq e^\epsilon \cdot \Pr[M_{EDLP}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Definition. (β -Smooth Upper Bound on EDLS for Private Selection)

For $\beta > 0$, a function $S : D^n \times \mathcal{R} \rightarrow \mathbb{R}$ is a β -smooth upper bound on the enhanced and direction-aware local sensitivity of a score function $u : D^n \times \mathcal{R} \rightarrow \mathbb{R}$ if it satisfies the following requirements for any $r \in \mathcal{R}$:

$$\begin{aligned} \forall x \in D^n : \quad & S(x, r) \geq EDLS_{u, \mathcal{R}}(x, r); \\ \forall x, y \in D^n, d(x, y) = 1 : \quad & S(x, r) \leq e^\beta \cdot S(y, r). \end{aligned}$$

Definition. (Enhanced and Direction-Aware β -Smooth Sensitivity for Private Selection)

For $\beta > 0$, the enhanced and direction-aware β -smooth sensitivity of a score function $u : D^n \times \mathcal{R} \rightarrow \mathbb{R}$ at x for $r \in \mathcal{R}$ is

$$EDS_{u, \mathcal{R}, \beta}^*(x, r) = \max_{y \in D^n} \left(EDLS_{u, \mathcal{R}}(y, r) \cdot e^{-\beta \cdot d(y, x)} \right).$$

Theorem 4. The EDSPS satisfies $\left(\left(k + \frac{|\mathcal{R}|}{2} \cdot l \right) \cdot \epsilon \right)$ -differential privacy.

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{EDSPS}(x) = 1] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{EDSPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x, r)$.

$$\begin{aligned} & \Pr[M_{EDSPS}(x) = 1] \\ &= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\ & \quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2))}{S(x)} \right] \\ & \quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m))}{S(x)} \right]. \end{aligned}$$

$$\begin{aligned} & \Pr[M_{EDSPS}(y) = 1] \\ &= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1))}{S(y)} \right] \\ & \quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(y, 2))}{S(y)} \right] \\ & \quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(y, m))}{S(y)} \right] \\ &\geq \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1))}{S(y)} \right] \\ & \quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2) - T')}{S(y)} \right] \\ & \quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m) - T')}{S(y)} \right] \end{aligned}$$

$$\begin{aligned} & \left(T' = \max_{r' \in \mathcal{R} \setminus \{1\}} (u(y, r') - u(x, r')) \right) \\ &= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T')}{S(y)} \right] \\ & \quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(y, 2))}{S(y)} \right] \\ & \quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(y, m))}{S(y)} \right]. \end{aligned}$$

Here, for all $v \in (-\infty, \infty)$

$$\begin{aligned} \forall r' \in \mathcal{R} \setminus \{1\} : \quad & \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(x)} \right] \\ & \leq e^{\frac{l\epsilon}{2}} \cdot \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(y)} \right]. \end{aligned}$$

In addition,

$$\begin{aligned} & \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\ & \leq e^{k\epsilon} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T')}{S(x)} \right] \end{aligned} \quad (2)$$

because

$$\begin{aligned} & \frac{|2\alpha'(u(x, 1) - u(y, 1) + T')|}{S(x)} \\ &= \frac{2\alpha' |u(x, 1) - u(y, 1) + \max_{r' \in \mathcal{R} \setminus \{1\}} (u(y, r') - u(x, r'))|}{S(x)} \\ &\leq \frac{2\alpha' \cdot EDLS_{u, \mathcal{R}}(x, 1)}{S(x)} \leq \frac{2\alpha' \cdot S(x, 1)}{S(x)} \leq 2\alpha' = \alpha(2k\epsilon), \end{aligned}$$

and

$$(2) \leq e^{k\epsilon} \cdot e^{\frac{l\epsilon}{2}} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T')}{S(y)} \right].$$

Therefore, $\Pr[M_{EDSPS}(x) = 1] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{EDSPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{EDSPS}(x) = r] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{EDSPS}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Theorem 5. The SPSPS satisfies $(k \cdot \epsilon + \max(\frac{l}{2} \cdot \epsilon, |\mathcal{R}| \cdot \beta'))$ -differential privacy when Z_r are sampled from the one-sided distribution g .

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{SPSPS}(x) = 1] \leq \exp(k \cdot \epsilon + \max(\frac{l}{2} \cdot \epsilon, m \cdot \beta')) \cdot \Pr[M_{SPSPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x, r)$.

$$\begin{aligned} & \Pr[M_{SPSPS}(x) = 1] \\ &= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{\alpha'(v - u(x, 1))}{S(x)} \right] \\ & \quad \cdot \Pr \left[Z_2 \leq \frac{\alpha'(v - u(x, 2))}{S(x)} \right] \\ & \quad \cdots \Pr \left[Z_m \leq \frac{\alpha'(v - u(x, m))}{S(x)} \right]. \end{aligned}$$

$$\begin{aligned} & \Pr[M_{SPSPS}(y) = 1] \\ &= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{\alpha'(v - u(y, 1))}{S(y)} \right] \\ & \quad \cdot \Pr \left[Z_2 \leq \frac{\alpha'(v - u(y, 2))}{S(y)} \right] \\ & \quad \cdots \Pr \left[Z_m \leq \frac{\alpha'(v - u(y, m))}{S(y)} \right] \end{aligned}$$

$$\begin{aligned}
&\geq \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{\alpha'(v - u(y, 1))}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{\alpha'(v - u(x, 2) - S(x))}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{\alpha'(v - u(x, m) - S(x))}{S(y)} \right] \\
&= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{\alpha'(v - u(y, 1) + S(x))}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{\alpha'(v - u(x, 2))}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{\alpha'(v - u(x, m))}{S(y)} \right].
\end{aligned}$$

(I) When $S(x) \geq S(y)$, for all $v \in (-\infty, \infty)$,

$$\begin{aligned}
\forall r' \in \mathcal{R} \setminus \{1\} : \quad &\Pr \left[Z_{r'} \leq \frac{\alpha'(v - u(x, r'))}{S(x)} \right] \\
&\leq \Pr \left[Z_{r'} \leq \frac{\alpha'(v - u(x, r'))}{S(y)} \right].
\end{aligned}$$

In addition,

$$\begin{aligned}
&\Pr \left[Z_1 = \frac{\alpha'(v - u(x, 1))}{S(x)} \right] \\
&\leq e^{k\epsilon} \cdot \Pr \left[Z_1 = \frac{\alpha'(v - u(y, 1) + S(x))}{S(x)} \right] \quad (3)
\end{aligned}$$

because

$$\begin{aligned}
0 &\leq \frac{\alpha'(u(x, 1) - u(y, 1) + S(x))}{S(x)} \leq \frac{\alpha' \cdot 2S(x)}{S(x)} \\
&= 2\alpha' = \alpha(2k\epsilon),
\end{aligned}$$

and

$$(3) \leq e^{k\epsilon} \cdot e^{\frac{k\epsilon}{2}} \cdot \Pr \left[Z_1 = \frac{\alpha'(v - u(y, 1) + S(x))}{S(y)} \right].$$

Therefore, $\Pr[M_{SPS}(x) = 1] \leq e^{k\epsilon + \frac{k\epsilon}{2}} \cdot \Pr[M_{SPS}(y) = 1]$.

(II) When $S(y) > S(x)$, note that when $\lambda < 0$, $\ln \left(\frac{h(z)}{e^{\lambda \cdot h(e^{\lambda \cdot z})}} \right) < |\lambda|$ for $z \geq 0$. Therefore, for all $v \in (-\infty, \infty)$,

$$\begin{aligned}
\forall r' \in \mathcal{R} \setminus \{1\} : \quad &\Pr \left[Z_{r'} \leq \frac{\alpha'(v - u(x, r'))}{S(x)} \right] \\
&\leq e^{\beta'} \cdot \Pr \left[Z_{r'} \leq \frac{\alpha'(v - u(x, r'))}{S(y)} \right].
\end{aligned}$$

In addition,

$$\begin{aligned}
&\Pr \left[Z_1 = \frac{\alpha'(v - u(x, 1))}{S(x)} \right] \\
&\leq e^{k\epsilon} \cdot e^{\beta'} \cdot \Pr \left[Z_1 = \frac{\alpha'(v - u(y, 1) + S(x))}{S(x)} \right].
\end{aligned}$$

Therefore, $\Pr[M_{SPS}(x) = 1] \leq e^{k\epsilon + m\beta'} \cdot \Pr[M_{SPS}(y) = 1]$. Consequently, $\Pr[M_{SPS}(x) = 1] \leq \exp(k \cdot \epsilon + \max(\frac{1}{2} \cdot \epsilon, m \cdot \beta') \cdot \epsilon) \cdot \Pr[M_{SPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{SPS}(x) = r] \leq \exp(k \cdot \epsilon + \max(\frac{1}{2} \cdot \epsilon, m \cdot \beta') \cdot \epsilon) \cdot \Pr[M_{SPS}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Theorem 6. M_{EDOEP} satisfies ϵ -differential privacy.

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{EDOEP}(x) = 1] \leq e^\epsilon \cdot \Pr[M_{EDOEP}(y) = 1]$. We denote $Z_r = \text{Expo}_r \left(\frac{\epsilon}{EDOGS_{u, \mathcal{R}}} \right)$.

$$\begin{aligned}
\Pr[M_{EDOEP}(x) = 1] &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(x, 1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x, 2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(x, m)].
\end{aligned}$$

$$\begin{aligned}
\Pr[M_{EDOEP}(y) = 1] &= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(y, 2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(y, m)] \\
&\geq \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x, 2) - T''] \\
&\quad \cdots \Pr[Z_m \leq v - u(x, m) - T''] \\
&\quad \left(T'' = \max_{r' \in \mathcal{R}} (u(y, r') - u(x, r')) \right) \\
&= \int_{v \in (-\infty, \infty)} \Pr[Z_1 = v - u(y, 1) + T''] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x, 2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(x, m)].
\end{aligned}$$

Here, $EDOGS_{u, \mathcal{R}}$

$$\begin{aligned}
&\geq u(x, 1) - u(y, 1) + T'' \\
&= -(u(y, 1) - u(x, 1)) + \max_{r' \in \mathcal{R}} (u(y, r') - u(x, r')) \\
&\geq -\max_{r \in \mathcal{R}} (u(y, r) - u(x, r)) + \max_{r' \in \mathcal{R}} (u(y, r') - u(x, r')) \\
&= 0.
\end{aligned}$$

Therefore, $\forall v : \Pr[Z_1 = v - u(x, 1)] \leq e^\epsilon \cdot \Pr[Z_1 = v - u(y, 1) + T'']$, and $\Pr[M_{EDOEP}(x) = 1] \leq e^\epsilon \cdot \Pr[M_{EDOEP}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{EDOEP}(x) = r] \leq e^\epsilon \cdot \Pr[M_{EDOEP}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Definition. (β -Smooth Upper Bound on EDOLS for Private Selection)

For $\beta > 0$, a function $S : D^n \times \mathcal{R} \rightarrow \mathbb{R}$ is a β -smooth upper bound on the enhanced and direction-aware local sensitivity for one-sided distribution cases of a score function $u : D^n \times \mathcal{R} \rightarrow \mathbb{R}$ if it satisfies the following requirements for any $r \in \mathcal{R}$:

$$\forall x \in D^n : \quad S(x, r) \geq EDOLS_{u, \mathcal{R}}(x, r);$$

$$\forall x, y \in D^n, d(x, y) = 1 : \quad S(x, r) \leq e^\beta \cdot S(y, r).$$

Definition. (Enhanced and Direction-Aware β -Smooth Sensitivity for Private Selection with One-Sided Distribution)

For $\beta > 0$, the enhanced and direction-aware β -smooth sensitivity for one-sided distribution cases of a score function $u : D^n \times \mathcal{R} \rightarrow \mathbb{R}$ at x for $r \in \mathcal{R}$ is

$$EDOS_{u, \mathcal{R}, \beta}^*(x, r) = \max_{y \in D^n} \left(EDOLS_{u, \mathcal{R}}(y, r) \cdot e^{-\beta \cdot d(y, x)} \right).$$

Theorem 7. The EDOSPS satisfies $(k \cdot \epsilon + \max(\frac{l}{2} \cdot \epsilon, |\mathcal{R}| \cdot \beta'))$ -differential privacy.

Proof. We let $\mathcal{R} = \{1, 2, \dots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{EDOSPS}(x) = 1] \leq \exp(k \cdot \epsilon + \max(\frac{l}{2} \cdot \epsilon, m \cdot \beta')) \cdot \Pr[M_{EDOSPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x, r)$.

$$\begin{aligned}
& \Pr[M_{EDOSPS}(x) = 1] \\
&= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2))}{S(x)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m))}{S(x)} \right] \\
&= \Pr[M_{EDOSPS}(y) = 1] \\
&= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1))}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(y, 2))}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(y, m))}{S(y)} \right] \\
&\geq \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1))}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2) - T'')}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m) - T'')}{S(y)} \right] \\
&\quad \left(T'' = \max_{r' \in \mathcal{R}} (u(y, r') - u(x, r')) \right) \\
&= \int_{v \in (-\infty, \infty)} \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T'')}{S(y)} \right] \\
&\quad \cdot \Pr \left[Z_2 \leq \frac{2\alpha'(v - u(x, 2))}{S(y)} \right] \\
&\quad \cdots \Pr \left[Z_m \leq \frac{2\alpha'(v - u(x, m))}{S(y)} \right].
\end{aligned}$$

(I) When $S(x) \geq S(y)$, for all $v \in (-\infty, \infty)$,

$$\begin{aligned}
\forall r' \in \mathcal{R} \setminus \{1\} : \quad & \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(x)} \right] \\
& \leq \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(y)} \right].
\end{aligned}$$

In addition,

$$\begin{aligned}
& \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\
& \leq e^{k\epsilon} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T'')}{S(x)} \right] \quad (4)
\end{aligned}$$

because

$$\begin{aligned}
0 & \leq \frac{2\alpha'(u(x, 1) - u(y, 1) + T'')}{S(x)} \leq \frac{2\alpha' \cdot EDOLS_{u, \mathcal{R}}}{S(x)} \\
& \leq 2\alpha' = \alpha(2k\epsilon),
\end{aligned}$$

$$\text{and (4)} \leq e^{k\epsilon} \cdot e^{\frac{l\epsilon}{2}} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T'')}{S(y)} \right].$$

Therefore, $\Pr[M_{EDOSPS}(x) = 1] \leq e^{k\epsilon + \frac{l\epsilon}{2}} \cdot \Pr[M_{EDOSPS}(y) = 1]$.

(II) When $S(y) > S(x)$, note that when $\lambda < 0$, $\ln \left(\frac{h(z)}{e^{\lambda \cdot h(e^{\lambda \cdot z})}} \right) < |\lambda|$ for $z \geq 0$. Therefore, for all $v \in (-\infty, \infty)$,

$$\begin{aligned}
\forall r' \in \mathcal{R} \setminus \{1\} : \quad & \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(x)} \right] \\
& \leq e^{\beta'} \cdot \Pr \left[Z_{r'} \leq \frac{2\alpha'(v - u(x, r'))}{S(y)} \right].
\end{aligned}$$

In addition,

$$\begin{aligned}
& \Pr \left[Z_1 = \frac{2\alpha'(v - u(x, 1))}{S(x)} \right] \\
& \leq e^{k\epsilon} \cdot e^{\beta'} \cdot \Pr \left[Z_1 = \frac{2\alpha'(v - u(y, 1) + T'')}{S(x)} \right].
\end{aligned}$$

Therefore, $\Pr[M_{EDOSPS}(x) = 1] \leq e^{k\epsilon + m\beta'} \cdot \Pr[M_{EDOSPS}(y) = 1]$.

Consequently, $\Pr[M_{EDOSPS}(x) = 1] \leq \exp(k \cdot \epsilon + \max(\frac{l}{2} \cdot \epsilon, m \cdot \beta')) \cdot \Pr[M_{EDOSPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{EDOSPS}(x) = r] \leq \exp(k \cdot \epsilon + \max(\frac{l}{2} \cdot \epsilon, m \cdot \beta')) \cdot \Pr[M_{EDOSPS}(y) = r]$ for all x and y such that $d(x, y) = 1$. $\square$

Theorem 8. Given a set of datasets $U \subseteq D^n$, for any $\beta (> 0)$ satisfying

$$\beta \leq \min_{x \notin U} \frac{1}{ud(x)} \cdot \ln \left(\frac{GS_f}{LS_f(x)} \right), \quad (5)$$

where $ud(x) := \min_{y \in U} d(y, x)$ represents the shortest distance between x and U , the following function S is a β -smooth upper bound:

$$S(x) = GS_f \cdot e^{-\beta \cdot ud(x)}.$$

Proof. From (5), for all $x \notin U$,

$$\beta \leq \frac{1}{ud(x)} \cdot \ln \left(\frac{GS_f}{LS_f(x)} \right) \iff LS_f(x) \leq GS_f \cdot e^{-\beta \cdot ud(x)}.$$

Therefore, it is sufficient to show that $\forall x, y, d(x, y) = 1 : S(x) \leq e^\beta \cdot S(y)$.

(I) When $x, y \in U$, $S(x) = S(y) = GS_f$.

(II) When $x \in U$ and $y \notin U$, $S(x) = GS_f$ and $S(y) = GS_f \cdot e^{-\beta}$ because $ud(y) = d(x, y) = 1$.

(III) When $x \notin U$ and $y \in U$, $S(x) = GS_f \cdot e^{-\beta}$ and $S(y) = GS_f$.

(IV) When $x \notin U$ and $y \notin U$, where X satisfies $X \in U$ and $d(X, x) = ud(x)$, and Y satisfies $Y \in U$ and $d(Y, y) = ud(y)$,

$$\begin{aligned}
e^\beta \cdot S(y) &= e^\beta \cdot GS_f \cdot e^{-\beta \cdot d(Y, y)} \\
&\geq e^\beta \cdot GS_f \cdot e^{-\beta \cdot d(X, y)} [\because d(Y, y) \leq d(X, y)] \\
&\geq e^\beta \cdot GS_f \cdot e^{-\beta \cdot (d(X, x) + d(x, y))} \\
&\quad [\because d(X, y) \leq d(X, x) + d(x, y)] \\
&= GS_f \cdot e^{-\beta \cdot d(X, x)} = S(x).
\end{aligned}$$

$\square$

II. COMPUTATION IN EXPERIMENTS

A. Computation of $d(x, y)$

To compute the Hamming distance between two datasets, we approximated it based on the values of b and c as follows:

$$d(T(b, c), T(b', c')) = \begin{cases} \left\lceil \frac{|(b+c)-(b'+c')|}{2} \right\rceil & ((b-b') \cdot (c-c') \geq 0) \\ \left\lceil \frac{\max\{|b-b'|, |c-c'|\}}{2} \right\rceil & ((b-b') \cdot (c-c') < 0) \end{cases},$$

where $T(b, c)$ represents a table that can be formed as the following table:

		Non-Transmitted Allele		Total
		A_1	A_2	
Transmitted Allele	A_1	a	b	$a + b$
	A_2	c	d	$c + d$
Total		$a + c$	$b + d$	$2n$

B. Computation of $ud(x)$

We used Theorem 8 to compute $S(x, r)$ for the proposed mechanisms using smooth sensitivity, M_{ESPS} , M_{EDSPS} , and M_{EDOSPS} . For these cases, we set U to $\{y \mid \exists r' \in \mathcal{R} \setminus \{r\}. \max_{z: d(y, z)=1} (|u(y, r) - u(z, r)| + |u(y, r') - u(z, r')|) > 12\}$, $\{y \mid \exists r' \in \mathcal{R} \setminus \{r\}. \max_{z: d(y, z)=1} |(u(y, r) - u(z, r)) + (u(z, r') - u(y, r'))| > 12\}$, and $\{y \mid \exists r' \in \mathcal{R}. \max_{z: d(y, z)=1} ((u(y, r) - u(z, r)) + (u(z, r') - u(y, r'))) > 12\}$, respectively. Here, we assume that the maximum changes in the values of $\max_{y: d(x, y)=1} |u(y, r) - u(x, r)|$, $\max_{y: d(x, y)=1} (u(y, r) - u(x, r))$, and $\max_{y: d(x, y)=1} (u(x, r) - u(y, r))$ when x is changed to z s.t. $d(z, x) = 1$ are almost $\frac{32}{b_r + c_r}$. Then, we approximated the value of $ud(x)$ for $x \notin U$ to

$$\begin{aligned} & \min_{r'} \left\lceil \frac{(12 - (ls_r + ls_{r'})) (b_r + c_r) (b_{r'} + c_{r'})}{32 \cdot (b_r + c_r + b_{r'} + c_{r'})} + 1 \right\rceil, \\ & \min_{r'} \left\lceil \frac{(12 - (ls_{1r} + ls_{2r'})) (b_r + c_r) (b_{r'} + c_{r'})}{32 \cdot (b_r + c_r + b_{r'} + c_{r'})} + 1 \right\rceil, \\ & \text{and } \min_{r'} \left\lceil \frac{(12 - (ls_{1r} + ls_{2r'})) (b_r + c_r) (b_{r'} + c_{r'})}{32 \cdot (b_r + c_r + b_{r'} + c_{r'})} + 1 \right\rceil, \end{aligned}$$

respectively, where $ls_r = \max_{y: d(x, y)=1} |u(x, r) - u(y, r)|$, $ls_{1r} = \max_{y: d(x, y)=1} (u(x, r) - u(y, r))$, and $ls_{2r} = \max_{y: d(x, y)=1} (u(y, r) - u(x, r))$.